LAB 6 REPORT

Code Coverage and Branch Coverage Analysis

Student Details

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Course: Software Engineering Test Automation (SETA)

Lab: 6

Application: Car Rental Rate and Fleet Telematics

Tools: Python 3.11.9, pytest 9.1.1, coverage.py 7.17.0

1. Introduction and Objectives

This lab tests the Car Rental Rate and Fleet Telematics application using pytest.
The objective is to exercise application statements and decision branches and measure statement and branch coverage using coverage.py.

2. Application Overview

process_rental_fleet validates rentals, skips unconfirmed bookings, selects Economy, SUV, or Luxury rates, calculates mileage charges, applies driver-age surcharges, counts risk flags, and returns STANDARD, EMPTY_BATCH, or HIGH_RISK_AUDIT.
check_fleet_telematics checks vehicle speed logs. It returns NO_DATA for empty input, GROUND_VEHICLE after two consecutive readings above 100, otherwise SAFE.

3. Test Design

Tests cover invalid arguments, confirmed/unconfirmed rentals, defaults, all vehicle rates, non-positive rental duration, negative mileage, the 300-mile boundary, long trips, young-driver surcharges, risk-audit thresholds, empty batches, and telematics speeding streaks/resets.

4. Execution Commands

```
python -m pip install -r requirements.txt
pytest -v
coverage erase
coverage run --branch -m pytest
coverage report -m
coverage html
```

5. Measured Results

Tests passed: 12 / 12 Tests failed: 0

Application statements: 51 Missed statements: 0

Application branches: 36 Partial branches: 0

Statement coverage: 100% Branch coverage: 100%

Total application module coverage: 100%

6. Conclusion

All 12 tests passed. The car_rental_rate.py application module achieved 100% statement coverage and 100% branch coverage. The measured report records 51 of 51 statements and 36 of 36 branches covered, with no missed statements or partial branches.

7. Evidence